Directed Study on Mapping Class Groups

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1 Abstract

Mapping class groups capture symmetries of spaces. The mapping class group $\mathrm{Mod}(S)$ of a surface S is the group of all 'deformed' symmetries of S. On one hand, it simplifies all symmetries of S by identifying neighboring ones, producing a discrete group; on the other hand, it preserves the intersection pattern of curves in S, offering an intuitive way to understand its nature. We studied their basic properties. The contents are mainly based on *A Primer on Mapping Class Groups* by Benson Farb and Dan Margalit[2].

2 Mapping Class Group Basics

Definition: Mapping Class Group

Let S be a connected and orientable surface with boundary of finite type. The **mapping class group** of S, denoted as $\mathrm{Mod}(S)$, is the isotopy class of all orientation-preserving homeomorphisms of S, where the homeomorphisms and isotopies are relative to the boundary components and punctures of S. It is a group by compositions.

Example: MCG of Simple surfaces

- $\mathrm{Mod}(\mathbb{D}^2)$ is trivial. This can be shown using the Alexander's trick to construct a concrete isotopy.
- $\mathrm{Mod}(\bar{\mathbb{A}}^2)$ is isomorphic to $\mathbb{Z}$. Identify $\bar{\mathbb{A}}^2$ with $\mathbb{S} \times [0,1]$ embedded in $\mathbb{R}^2$. Let $T : \bar{\mathbb{A}}^2 \to \bar{\mathbb{A}}^2$ be given by the formula $T(\theta, t) = (\theta + 2\pi t, t)$. The isotopy class of T generates $\mathrm{Mod}(\bar{\mathbb{A}}^2)$.
- $\mathrm{Mod}(\mathbb{T}^2)$ is isomorphic to $\mathrm{SL}_2(\mathbb{Z})$.

3 Dehn twists

Dehn twists are the simplest infinite order elements in the mapping class groups, and play a fundamental role. The mapping class group of the compact annulus is generated by a 'twist' map $[T]$ as stated above. The idea of the Dehn twists is that we can do this twist map on a small subsurface on any surface. We define Dehn twists as follows:

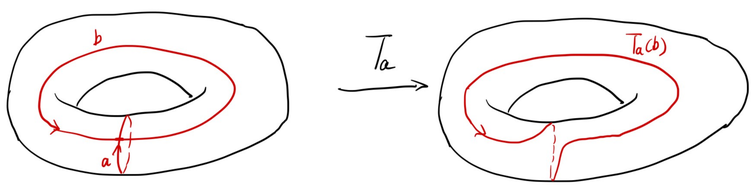
Definition: Dehn Twist

Let S be a connected and orientable surface with boundary of finite type, $[\alpha]$ be a simple loop in S, $\bar{U}$ be a compact tubular neighborhood of α in S, and $\phi : \bar{\mathbb{A}}^2 \to \bar{U}$ be an orientation-preserving homeomorphism. Define the Dehn twist T_α along α as:

$$T_\alpha(x) = \begin{cases} \phi \circ T \circ \phi^{-1}(x) & \text{if } x \in \bar{U} \\ x & \text{if } x \notin \bar{U} \end{cases}$$

It can be shown that the isotopy class of T_α is independent of the choice of $\alpha, \bar{U}, \phi$, and is well-defined.

It's often useful to consider Dehn twists' action on simple closed curves:



We can use this action to prove many good properties of Dehn twists. Let a and b be two isotopy classes of essential simple closed curves. Then

- T_a has infinite order;
- $T_a = T_b$ if and only if $a = b$;
- $T_a T_b T_a = T_b T_a T_b$ when $i(a,b) = 1$ (Braid relation);
- T_a, T_b has no relation when $i(a,b) \geq 2$.

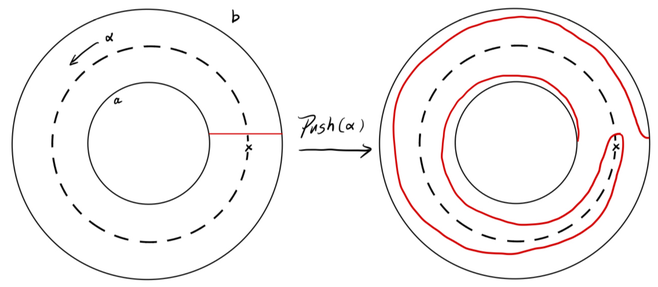
4 Birman Exact Sequence

Theorem: Birman Exact Sequence

Let S be a hyperbolic surface, and (S, x) be the surface obtained from S by marking a point x in the interior. Then the following sequence is exact:

$$0 \longrightarrow \pi_1(S,x) \xrightarrow{Push} \mathrm{Mod}(S,x) \xrightarrow{Forget} \mathrm{Mod}(S) \longrightarrow 0$$

Proof 1. We first construct the *Push* map for simple loops in $\pi_1(S,x)$ as illustrated below.



Then we notice that the *Push* map actually factors through the automorphism group of $\pi_1(S)$ as shown below.

$$\begin{array}{ccccc} & & \mathrm{Mod}(S,x) & \xrightarrow{Forget} & \mathrm{Mod}(S) \\ & & \downarrow \eta & & \downarrow \\ \{e\} & \longrightarrow & \mathrm{Inn}(\pi_1(S,x)) & \xrightarrow{\iota} & \mathrm{Aut}(\pi_1(S,x)) \longrightarrow \mathrm{Out}(\pi_1(S,x)) \longrightarrow \{e\} \end{array}$$

Then we show that the map η is injective, and we use the simple loops to 'build' the *Push* map for every element in $\pi_1(S,x)$ making the following diagram commutes:

$$\begin{array}{ccc} \pi_1(S,x) & \overset{Push}{\dashrightarrow} & \mathrm{Mod}(S,x) \\ \downarrow \cong & & \downarrow \eta, \text{injective} \\ \mathrm{Inn}(\pi_1(S,x)) & \xrightarrow{\iota} & \mathrm{Aut}(\pi_1(S,x)) \end{array}$$

Details can be found in [3]. □

Proof 2. There exists a fibre bundle as follows:

$$\mathrm{Homeo}^+(S,x) \longrightarrow \mathrm{Homeo}^+(S) \xrightarrow{\varepsilon} S$$

The theorem now follows from the long exact sequence of homotopy groups associated with a fibre bundle. This is Birman's original proof, and details can be found in [1] and [2]. □

The Birman exact sequence gives important insights into how the mapping class group is affected when marking a point. It is used in showing the following important property of mapping class groups as well as finite generation-ness.

Theorem: Centerless Theorem

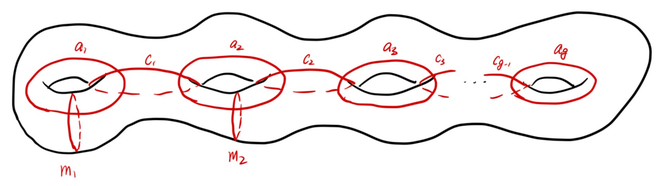
Let $S_{g,n}$ denotes the surface with genus g and n punctures. Then $\mathrm{Mod}(S_{g,n})$ is centerless except for $(g,n) = (0,2), (1,0), (1,1), (1,2), (2,0)$.

5 Finite generation

Theorem: Finite Generation

Let S_g denotes the surface with genus g. Then $\mathrm{Mod}(S_g)$ is finitely generated by Dehn twists along non-separating simple closed curves.

The proof mainly uses induction on genus and puncture where we consider the effect on the mapping class group when cutting a surface and filling a puncture. Some machinery from geometric group theory is also applied. By further refining the proof, it can be shown that S_g is finitly generated by the $2g+1$ Dehn twists below. The number $2g+1$ cannot be further optimized. Details can be found in [2].



References

[1] Joan Birman. Mapping class groups and their relationship to braid groups. *Communications on Pure and Applied Mathematics*, 22(2):213–238, 1969.

[2] Benson Farb and Dan Margalit. *A Primer on Mapping Class Groups*, volume 49 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 2012.

[3] Henry Wilton. Mapping class groups, 2021.